

Heart Disease Risk Prediction Model Using Machine Learning

Abstract

Heart disease is one of the leading causes of mortality worldwide. Early identification of individuals at risk can significantly improve treatment outcomes and reduce healthcare costs. This project presents a Machine Learning-based Heart Disease Risk Prediction Model designed to predict the likelihood of heart disease using patient medical data. The project demonstrates the complete machine learning pipeline, including data preprocessing, exploratory data analysis, model training, evaluation, and prediction. Multiple classification algorithms were implemented and compared to determine the most effective model for heart disease prediction. The developed system can assist healthcare professionals in making data-driven decisions for early diagnosis and preventive care.

Introduction

Cardiovascular diseases remain a major global health challenge. Traditional diagnosis often involves extensive clinical testing and expert evaluation, which may be time-consuming and costly. Machine learning techniques provide an efficient solution by analyzing patient health records and identifying patterns associated with heart disease risk.

The primary goal of this project is to develop a predictive model capable of classifying patients as either at risk or not at risk of heart disease based on various clinical attributes.

Objectives

- Develop a machine learning model for heart disease risk prediction.
 - Analyze medical attributes associated with cardiovascular diseases.
 - Compare the performance of multiple classification algorithms.
 - Evaluate model effectiveness using standard machine learning metrics.
 - Demonstrate the application of predictive analytics in healthcare.
-

Dataset Description

The project utilizes a heart disease dataset containing patient medical information and a target variable indicating the presence or absence of heart disease. The dataset includes several clinical features such as:

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol Level

- Fasting Blood Sugar
- Resting ECG Results
- Maximum Heart Rate Achieved
- Exercise-Induced Angina
- ST Depression (Oldpeak)
- Slope of Peak Exercise ST Segment
- Number of Major Vessels
- Thalassemia

The target variable represents whether the patient has heart disease.

Methodology

1. Data Collection and Preparation

Before model development, several preprocessing steps were performed:

- Data cleaning and validation
- Handling missing values
- Feature selection
- Data transformation and encoding
- Exploratory Data Analysis (EDA)
- Splitting the dataset into training and testing sets

2. Exploratory Data Analysis

EDA was conducted to understand feature distributions, identify correlations, and detect patterns associated with heart disease occurrence. Visualization techniques were used to gain insights into patient characteristics and disease prevalence.

3. Machine Learning Models

The following classification algorithms were implemented:

Logistic Regression

A statistical classification technique commonly used for binary prediction problems.

Support Vector Machine (SVM)

A supervised learning algorithm effective in high-dimensional classification tasks.

Random Forest Classifier

An ensemble learning method that combines multiple decision trees to improve prediction performance.

K-Nearest Neighbors (KNN)

A distance-based classification algorithm that predicts outcomes using neighboring data points.

Gradient Boosting Classifier

An ensemble technique that builds models sequentially to minimize prediction errors.

Evaluation Metrics

To evaluate model performance, the following metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix

These metrics provide a comprehensive assessment of model reliability and predictive capability.

Results and Discussion

The performance of different machine learning models was compared using the testing dataset.

Model	Accuracy
Logistic Regression	76.09%
Support Vector Machine (SVM)	78.26%
Random Forest	76.09%
K-Nearest Neighbors (KNN)	73.91%
Gradient Boosting	73.91%

Among all implemented models, the Support Vector Machine (SVM) achieved the highest prediction accuracy of **78.26%**, making it the best-performing model for this dataset.

The ROC-AUC scores also demonstrated strong predictive capability across all models, indicating that machine learning can effectively assist in identifying individuals at risk of heart disease.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Jupyter Notebook

Key Features

- Data preprocessing and cleaning
- Exploratory Data Analysis (EDA)
- Multiple machine learning model implementation
- Model comparison and performance evaluation
- Heart disease risk prediction for new patient data
- User input-based prediction system

Conclusion

This project successfully demonstrates the application of machine learning techniques in heart disease risk prediction. By leveraging patient medical data and classification algorithms, the developed system provides an efficient and data-driven approach for identifying potential heart disease cases.

Comparative analysis revealed that the Support Vector Machine (SVM) model achieved the best overall performance with an accuracy of approximately 78.26%. The project highlights important machine learning concepts including data preprocessing, feature engineering, model training, evaluation, and healthcare analytics.

The developed prediction model can serve as a valuable decision-support tool for healthcare professionals and contribute to early diagnosis and preventive healthcare strategies.

Future Enhancements

- Perform hyperparameter tuning to improve model accuracy.
- Implement advanced ensemble methods such as XGBoost and LightGBM.
- Apply feature selection techniques for improved efficiency.
- Deploy the model as a web application using Flask or Streamlit.
- Add real-time patient risk prediction capabilities.
- Integrate larger healthcare datasets for better model generalization.
- Develop an interactive dashboard for healthcare professionals.

References

1. Scikit-learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. UCI Heart Disease Dataset
5. Machine Learning for Healthcare Research Literature
6. Matplotlib Documentation
7. Seaborn Documentation